



Black-Box Entropy Estimation

Midterm Report

Technical Director

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University Project Supervisors

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Project Summary

Description

The Random number generator is a fundamental and important cryptographic element, which is commonly used and requires a random seed from an entropy source in order to output random data. However, in reality, the random number generator cannot provide enough randomness as expected which exposes cryptographic applications to vulnerabilities within the system to outside actors. It is important and necessary to determine that the seeds are sufficiently random. This can be solved by using Neural Networks (black box) in a sense that while it can approximate any function, studying its structure won't give you any insights on the structure of the function being approximated. ⁽¹⁾

Team Responsibilities

- Learn about entropy estimation and become familiar with pseudorandom number generators (PRNG)
- Understand RSA and DSA public-key cryptosystems and weaknesses that could compromise private keys
- Become familiar with Machine learning techniques used with Tensorflow and Neural Networks
- Derive entropy estimators and test them

Budget

We do not expect any financial costs to this project. We expect to collectively work 8-10 hours per week on this project and meet at least twice per week.

Deliverables

- Proposal and Literature Review
- Midterm Presentation
- Final Presentation

Broader Impact

The scope of the project is to provide an extensive application to the resolution of how randomness can affect the securities, vulnerability of applications, especially in cryptography. Randomness has and always will play an important part of encryption which is used in many aspects of life and technologies (mobile, telecom, data, and

security,...). In solving and finding new ways to explore the problem, we hope such contributions will help to develop a baseline to the future development of encryption.

Plan

Meetings will be scheduled to meet and be consulted by the sponsor on every Friday of the week.

Tentative Schedule

September 10, 2021	Submit Proposal
September 17, 2021	Research and explore topics
September 24, 2021	Set up environment and experiment
October 24, 2021	Midterm Report
December 9, 2021	Final Report

Recurring meeting schedule

Friday 1 P.M - 3:45 P.M
Team meets regularly on Tuesday and Thursday.

Midterm Review

Project Approach

In cryptography, randomness in systems is crucial to ensure security and a concrete design. It's used in key generation, padding, nonces, initialization vectors, and much more. A sequence of numbers is random if it is unpredictable; as in knowing previous outputs of a sequence does not help in predicting future outputs, and knowing a sequence is random for certain is impossible. So, we look for a method to approximate the unpredictability of a noise source.

The main goal of this project is to understand how random number generators work and to learn about machine learning and how important machine learning can be when it comes to the evaluation of how good the unpredictability is when it comes to generating said random numbers. After a few weeks of research, our team is looking at approaching the project using two known supervised machine learning techniques: linear regression and logistic regression.

Detailed Plan

Project Timeline

SEPT 10

began research and exploration

- ❖ min-entropy, pseudo-random number generators, etc.

SEPT 24

set up environment

- ❖ began setting up in Colaboratory and experimenting with Tensorflow
- ❖ imported models from other projects and topics

OCT 8

prepared data sets for model

- ❖ derived data values from varying types of distributions
- ❖ imported data into Google Colaboratory
- ❖ analyzing the data sets

OCT 22

preparing approach to analyze data through logistic regression

- ❖ began by looking at linear regression
- ❖ later switching to logistic regression approach

NOV 5

analyze and experiment through logistic regression

- ❖ analyze the model and receive results from logistic regression with imported data
- ❖ test using different data sets and multiple results
- ❖ Plot data for visualisation.

DEC 3

analyze results

- ❖ analyze results and derive conclusions
- ❖ look to future improvements and possibilities

Work in Progress

- ❖ Google colab instance was set up and used to run numeric distribution data (CSV files with 1,000,000 rows of binary data)
- ❖ Starting out with a truncated Binomial distribution file with only 10,000 (collab resource limitation)
- ❖ CSV file taken and dissected into 4 independent (X1, X2, X3, X4) and 1 dependent (Y) variables and put into a python dataframe (500 X 5)

Next Plan

- ❖ All columns in dataframe created from to be converted into integer values and used to experiment with linear and logistic regression techniques
- ❖ Generate basic accuracy metrics (Using a confusion matrix)
- ❖ Produce a receiver operating characteristic (ROC) for visualization (AUC..etc.)

Challenges

There were a few challenges faced by the team. One was having a clear understanding of the project's objective, but that was clarified by our technical director in multiple learning sessions during our weekly team meetings. Another challenge is that most of our team is completely new to machine learning and its applications. In order to combat that, our team would watch multiple online tutorials based on the topics of machine learning and its applications and then a secondary team meeting would be held to discuss what was learned and how it would be applicable to our project and its objectives.

Literature Review

Predictive Models for Min-Entropy Estimation

John Kelsey, Kerry A. McKay, and Meltem Sonmez Turan

This paper models the use of predictors to estimate min-entropy to successfully predict a source's outputs. It begins by introducing a cryptographic pseudorandom number generator (PRNG) and entropy estimation as well as fundamental entropy definitions, and further underscores the need to investigate the procedures and relations between entropy per sample from entropy source. The paper explains various predictors and its general model that provides a prediction and updates its internal state based on the sample in order to strengthen future predictions, ultimately producing training and testing sets. The paper applied the predictors to real-world data, matching the approach and value of min-entropy computation for 90B estimates. The resulting estimator for a set of 90B estimators was capped at six bits of entropy per sample. This work will be used in our research and future explorations in this project to better understand the predictor model and advance towards more machine-learning tools.

High-Efficiency Min-Entropy Estimation Based on Neural Network for Random Number Generators

Na Lv, Tianyu Chen, Shuangyi Zhu, Jing Yang, Yuan Ma, Jiwu Jing, Jingqiang Lin, and Clemente Galdi

The paper emphasizes the critical role that random number generators play in securing digital information as a critical component of cryptographic algorithms. For this

reason, the output of a RNG must be truly random and unpredictable. The challenge is then determining a satisfactory level of uncertainty or entropy to be achieved by a random number generator. The paper discusses the application of neural networks to this problem of min-entropy and proposes an approach that has applications to not only stationary sources but also non stationary ones. The discussion of a machine learning-based approach to the problem at hand, specifically neural networks, may provide a solid starting point for the research.

Linear Feedback Shift Register

Texas Instruments - John Koeter

This paper illustrates the use of bit shifters to generate a random number which played an essential role in the past in telecom, cryptography and computer encryption. The simple explanation of this paper is that it uses the shifting method of bit in the number to shift the bit of number from top of queue to the bottom and use a logic gate (XOR) to generate a new bit to put on the top. This method is an extremely good pseudo-random pattern generator. However, in some cases, we want to increase the complexity of the function (linear) to some degree of polynomial to increase the states and hence making it harder to decipher.

Deep Learning-Based Security Verification for a Random Number Generator Using White Chaos

Cai Li, Jianguo Zhang, Luxiao Sang, Lishuang Gong, Longsheng Wang, Anbang Wang, and Yuncai Wang

This paper proposes a deep learning (DL)-based predictive analysis of the security of a non-deterministic random number generator (NRNG) that uses a white noise entropy based source. Particularly, a temporal pattern attention (TPA)-based DL model is created in which it analyzes the data from two different stages of a NRNG which are the output of a chaotic external-cavity semiconductor laser (ECL) and the final output from the NRNG itself. The model processes and learns from this data and is able to produce promising results according to the authors.

Investigator Biographies

Omar Abbas

Omar Abbas graduated with a BS degree in Computer Engineering in 2019 and is currently in his last semester completing the Fast-track MS in Computer Engineering program at UT Dallas. Throughout his undergraduate and graduate studies, Omar has taken courses in information security, database design, operating systems and web programming languages. He is currently working for T-Mobile as a 3rd level support and deployment Engineer. He has experience with Java, C++, Python, and Bash. He primarily utilizes his skills to facilitate, optimize, and automate deployments in T-Mobile's Charging System.

Daniel Le

Daniel is a senior undergraduate student currently doing a fast-track Master of Science in Computer Science at University of Texas at Dallas. He was an intern at Tyler Technologies, and also possess website development skills as well as knowledge in programming languages such as Java, Python, C++, and MySQL. He has taken courses in Machine Learning, Data Science and is looking to utilize those skills in future projects.

Alinta Wang

Alinta has recently graduated with a Bachelor's of Science in Computer Science this past Spring at the University of Texas at Dallas, and is looking to pursue her Master's degree through Fast Track with a focus in Cybersecurity at UTD as well. She has taken courses in Information Security, Network Security, and Big Data Security and Privacy. She is familiar with coding in languages such as Java, SQL, C, and shell. She has interned at Intel Security and recently finished an internship at Fidelity Investments as well, and hopes to advance and utilize her knowledge in future projects.

Gabriel Goetten

Gabriel graduated with a BS in Computer Engineering in 2020 and is currently in his last semester of graduate studies in Information Assurance. Throughout his studies, Gabriel has taken courses in information security, computer networks and network security, as well as malicious code analysis. He also possesses software development skills with a few general purpose programming languages such as C++ and JavaScript. He is currently an intern at a financial technology startup, where he is part of the information security team, working with access control and security policy development.

References

Kelsey, John, McKay, Kerry A, Turan, Meltem Sonmez Turan. "Predictive Models for Min-Entropy Estimation." 2015.

Na Lv, Tianyu Chen, Shuangyi Zhu, Jing Yang, Yuan Ma, Jiwu Jing, Jingqiang Lin, and Clemente Galdi. "High-Efficiency Min-Entropy Estimation Based on Neural Network for Random Number Generators." 2020.

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Additional References

⁽¹⁾ R. Hewage, "Deep learning & artificial neural Networks: Solving the black Box Mystery," *Hacker Noon*. [Online]. Available: <https://hackernoon.com/deep-learning-and-artificial-neural-networks-solving-the-black-box-mystery-rl7h32wa>. [Accessed: 30-Aug-2021].

⁽²⁾ M. Li, K. Ma, J. You, D. Zhang and W. Zuo, "Efficient and Effective Context-Based Convolutional Entropy Modeling for Image Compression," in *IEEE Transactions on Image Processing*, vol. 29, pp. 5900-5911, 2020, doi: 10.1109/TIP.2020.2985225.